



## Module 11 Day 2: Cryptography

### Solution Guide: Generating Hashes

The goal of this exercise was to use hashes to validate the integrity of data. You were tasked with generating MD5 hashes to determine which files were modified.

- Extract the investigation summary files.
- Navigate to the backup directory:
  - Run `cd /Alphabet_Bandit_Investigation_Reports/backup`.
- Create a hash for each of the files in the directory and save this to a single file:
  - `md5sum * > hashes`
- Preview the hash file:
  - `more hashes`
- The file should look like:

Unset

```
94d53dff0ba90e465fcf1e7b900ec843 *Investigation_1101
abf9816ba8e141b301848e09038ed1f8 *Investigation_1108
3edab292de508c37fc8aeb5fd973d54b *Investigation_1110
609c6b71a2241e1b8f878c682d138c2d *Investigation_1112
2147a20472c421f2b9af8250d742bbd1 *Investigation_1116
```

- Move the hash file to the current folder:
  - `mv hashes ../current/`
- Navigate to the current folder:
  - `cd ..`
  - `cd current`
- Run the following command to compare the hashes from the backup folder and the current folder:
  - `md5sum -c hashes`
- The results should clearly show the two files are different (Investigation\_1101 and Investigation\_1110):

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```
Investigation_1101: FAILED
Investigation_1108: OK
Investigation_1110: FAILED
Investigation_1112: OK
Investigation_1116: OK
md5sum: WARNING: 2 computed checksums did NOT match
```

- Next, since we know which two files were modified, run the following diff commands to find out what was changed:
  - `diff Investigation_1101 ../backup/Investigation_1101`
  - `diff Investigation_1110 ../backup/Investigation_1110`
- The results should show:
  - 1101:

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```
< Dr Browns Residence was burglarized, exotic car was taken
as well as several containers of Plutonium.
---
> Dr Browns Residence was burglarized, exotic car was taken
as well as several containers of Gasoline
```

- 1110

Unset

```
< Jennifer Parkers House was burglarized, mostly jewelry
was stolen and significant damage occurred at the house
---
> Jennifer Parkers House was burglarized, mostly jewelry
was stolen and no damage occurred at the house
```

- These outputs show the following changes:
  - In the 1101 file, the word Plutonium was changed to Gasoline.
  - In the 1110 file, the word significant was changed to no.